

Priority & compare

When several requests arrive at once, a priority encoder picks one winner and reports its index

Winner, flags, cascade

- High-index-first: if I zero and I two are both set, the winner is two
- Low-index-first would pick zero instead, priority is a design choice
- With all inputs zero and enable-in high
- Compare four-bit A and B
- Same bits can yield A greater unsigned but A less signed

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Priority encoder & comparator

Resolve which request wins under high- or low-index priority (with enable cascade), then compare A and B as unsigned vs two's-complement signed — including cases where the flags disagree.

Starter example: 4-input priority encoder, high-index-first, I0=I2=1 → winner I2 (Y=2).

Challenge 0 / 22 cleared

High-pri: I2 wins: 4-input, high-index-first: set I2=1 and I0=1; winner must be 2.

Show hint

Check

Next

Load challenge setup

Idle

High-pri: I2 wins

Low-pri: I0 wins

No request

EI disabled

Encode I3

8-in: I6 wins

8-in low priority

Quiz: E0 meaning

Quiz: priority

Compare equal

Unsigned A>B

Unsigned A<B

Signed -1 vs 1

Flags disagree

Signed A>B

8-bit 0xFF vs 0x01

8-bit equal

Quiz: signed MSB

Quiz: unsigned 1000

Quiz: when flags differ

Quiz: cascade use

Both zero

Lab

Load starter example

BLOCK

Priority encoder

INPUTS

4 → 2

PRIORITY

High-index first

Notes

Y = max { i | Ii=1 } when EI=1; else idle

Winner I2 → Y=I0₂ (2)

Priority compare starter

Workbook practice

- In the workbook track
- With enable-in high and no inputs active, state V and enable-out
- For four-bit A equals fifteen and B equals one, say unsigned and signed relational results
- Name one pitfall: assuming comparator flags match without checking signed mode

Pitfalls to watch

- Do not mix up priority direction, high versus low index first changes the answer
- Enable-out is not “any input high”; it means no local winner while enable-in is on
- And remember: the browser lab is literacy
- Real buses still need timing, width agreement

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, sketch one priority case and one compare disagree case
- When you are ready, take the short quiz, then continue to half and full adder